Part E: Mortgage Survival Analysis

Competing Risks · Time-Varying Cox · Neural Models · Scenario Analysis

Yueqi Lin

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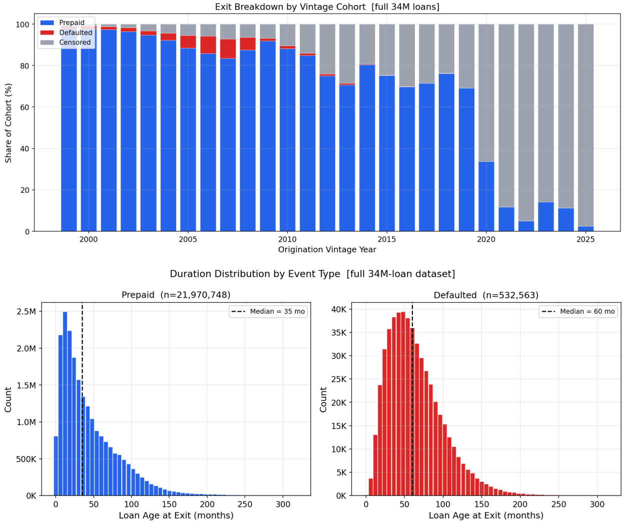
Part E(i)

Competing Risks: Prepayment vs. Default

Kaplan–Meier vs. Aalen–Johansen · Cause-Specific Cox · Fine–Gray

Prepaid and default are competing risks.



Kaplan–Meier vs. Aalen–Johansen.

Kaplan–Meier

$$\hat{F}^{(k)}_{\mathrm{KM}}(t) = 1 - \prod_{t_j \le t}\left(1 - \frac{d_j^{(k)}}{n_j^{\mathrm{KM}}}\right)$$

$d_j^{(k)}$	cause-k exits at t_j
n_j^{KM}	**risk set excluding competing-event loans**

Aalen–Johansen

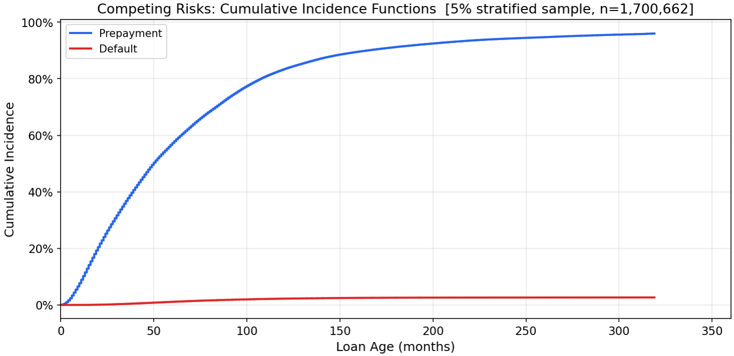
$$\hat{F}_k(t) = \sum_{t_j \le t} \hat{S}(t_j^-) \cdot \frac{d_{kj}}{n_j}$$

d_{kj}	cause-k exits at t_j
n_j	risk set including all exits
$\hat{S}(t_j^-)$	**overall survival just before t_j**

KM: competing exits censored $\Rightarrow$ denominator too small $\Rightarrow$ each increment inflated.

AJ: $\hat{S}(t_j^-)$ shrinks with every exit $\Rightarrow$ $\hat{F}^{(\mathrm{pre})} + \hat{F}^{(\mathrm{def})} + \hat{S} = 1$ always.

Aalen–Johansen CIF: coherent probability partition.



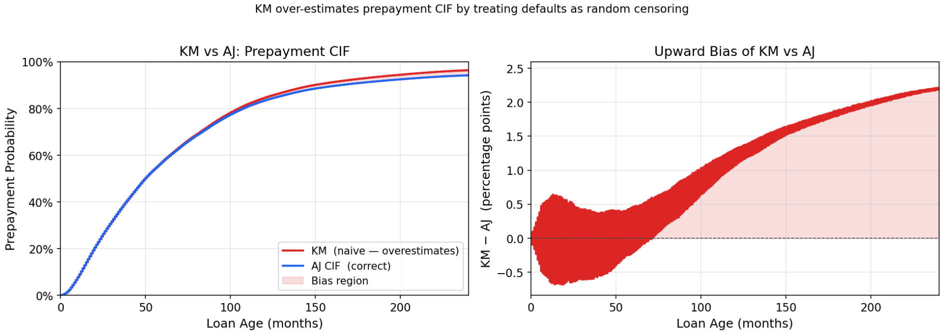
Cumulative Incidence Function

$$F_k(t) = P(\text{exit by } t,\ \text{cause } k)$$

Aalen–Johansen CIF partition

	Prepaid	Default	Active
5 yr	61.4%	1.1%	37.6%
10 yr	83.3%	2.2%	14.5%
20 yr	91.8%	3.1%	5.1%

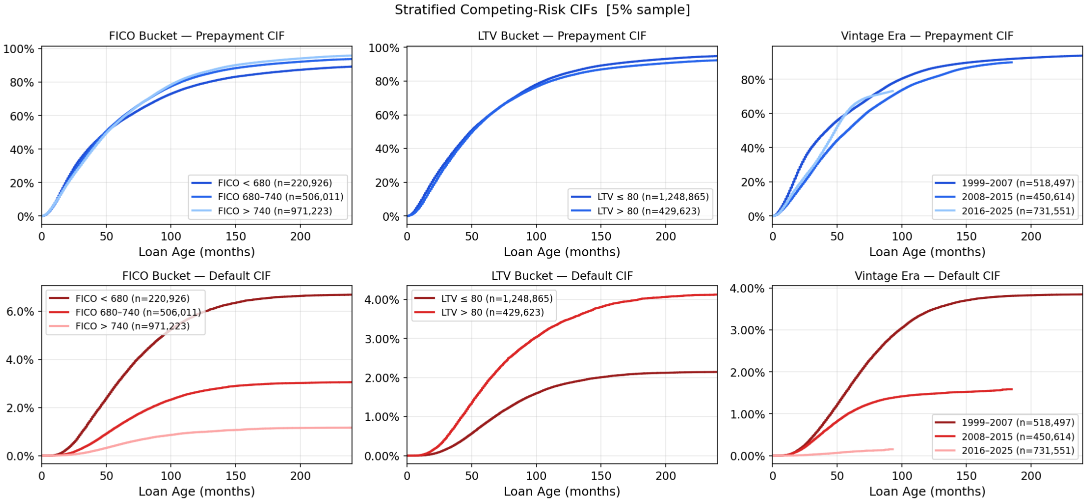
KM overstates prepayment by up to 2.2 pp.



KM overstatement of prepayment probability

Horizon	$\hat{F}_{\mathrm{KM}}$	$\hat{F}_{\mathrm{AJ}}$	Bias
5 yr	61.9%	61.4%	+0.45 pp
10 yr	84.6%	83.3%	+1.31 pp
20 yr	93.4%	91.8%	+2.23 pp

FICO and LTV separate default, not prepayment.



Cause-specific Cox model.

Cause-Specific Cox

$$h_k(t \mid \mathrm{x}) = h_{0k}(t)\,\exp\!\left(\boldsymbol{\beta}_k^\top \mathrm{x}\right), \quad k \in \{\text{pre}, \text{def}\}$$

$$\ell(\boldsymbol{\beta}_k) = \sum_{i:\,E_i=k} \left[\boldsymbol{\beta}_k^\top \mathrm{x}_i - \log \sum_{j\in\mathcal{R}(t_i)} e^{\boldsymbol{\beta}_k^\top \mathrm{x}_j}\right]$$

$\mathcal{R}(t_i)$: all loans still active just before t_i.

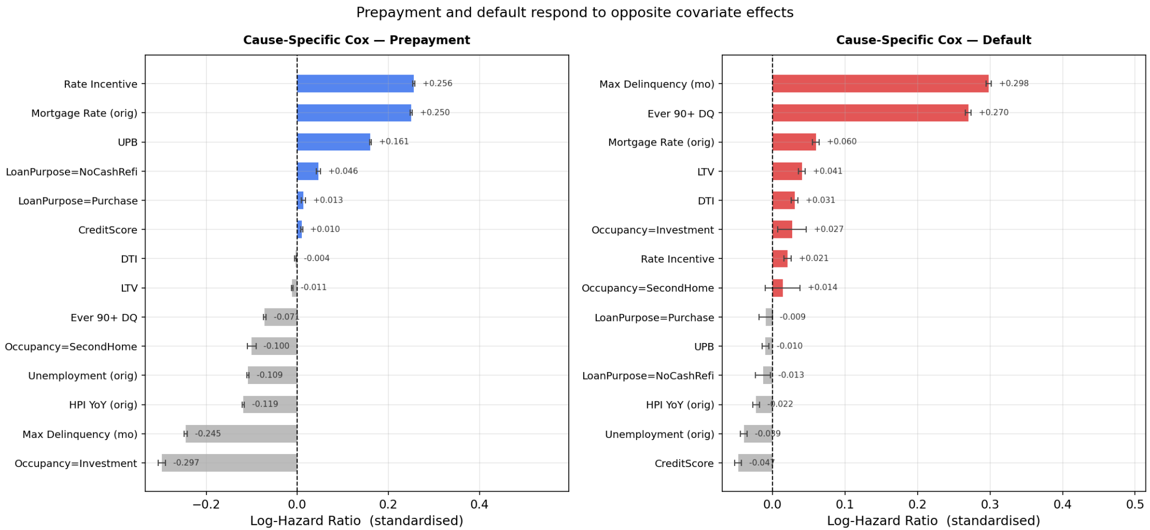
Competing-cause exits are **censored** — they leave $\mathcal{R}(t)$ without contributing to ℓ.

Additional features

Max Delinquency — months past due over loan life

Ever 90+ DQ — binary flag for severe delinquency

Opposite covariate effects by cause.



Four covariates flip sign across causes.

Naive Cox vs. cause-specific coefficients

Covariate	**Naive $\hat{\beta}$**	**Prepay $\hat{\beta}$**	**Default $\hat{\beta}$**	
Rate incentive	$+0.253$	$+0.256$	$+0.021$	(pure prepay)
UPB	$+0.161$	$+0.161$	-0.010	(pure prepay)
FICO	$+0.016$	$+0.010$	-0.047	**opposite**
LTV	-0.007	-0.011	$+0.041$	**opposite**
Max Delinquency	-0.123	-0.245	$+0.298$	**opposite**
Ever 90+ DQ	-0.035	-0.071	$+0.270$	**opposite**

Each extra month delinquent raises default hazard by 35% and cuts prepayment hazard by 22% — yet the naive coefficient (-0.123) conceals both, collapsing a $+0.543$ spread to near zero.

Key Takeaways: E(i)

1. Default and prepayment are competing risks — a naive Cox model conflates both.
2. **Kaplan–Meier overstates prepayment by up to $+2.2$ pp; use Aalen–Johansen.**
3. Vintage era dominates: 2007 cohort defaults at $6\times$ the long-run average.
4. FICO drives default; rate incentive drives prepayment — separate channels.

Part E(ii)

Time-Varying Covariates in Mortgage Survival

Andersen–Gill Cox Extension · Live Rate Incentive · Landmark C-index

Static Cox vs. Andersen–Gill.

Standard Cox (11 features)

One row per loan; all covariates fixed at origination. Cannot track rate cycles, burnout, or equity build after origination.

Andersen–Gill (13 features)

One row per loan-month; $x_i(t)$ updated each period. Extends the same static base with the four time-varying features, tracking the full path of rate incentive and equity. 100K loans $\times$ $\sim$47 months $=$ 4.68M intervals.

Time-varying features — updated every month

Feature	**What it captures**
`rate_incentive`	$r_{\text{orig}} - r_{\text{mkt}}(t)$: live option value to refinance; tracks rate cycles and burnout
`ELTV`	equity from amortisation and HPI gains; high ELTV suppresses refi
`unemployment`	macro labour cycle; proxies Fed easing in high-unemp periods
`hpi_yoy`	home-price appreciation unlocking cash-out refinancing

*`unemployment` and `hpi_yoy` also in Standard Cox as origination-month snapshots.

The Andersen–Gill model.

Hazard function (identical in form to standard Cox)

$$h_i(t) \;=\; h_0(t)\,\exp\!\big(\boldsymbol{\beta}^\top \mathrm{x}_i(t)\big)$$

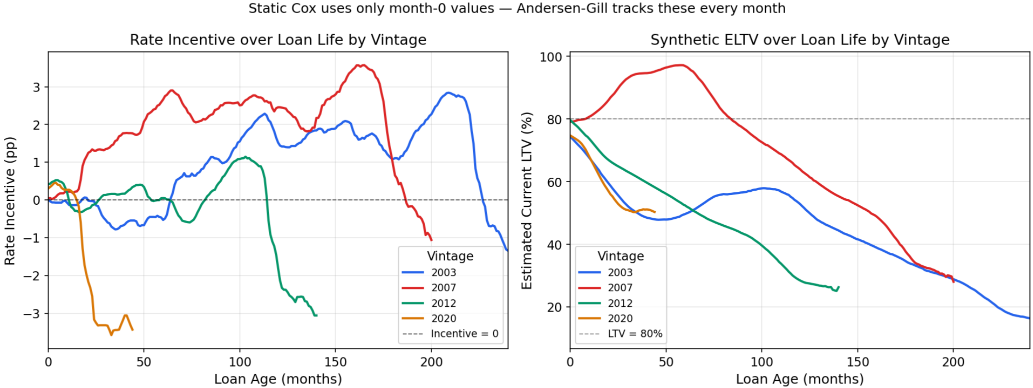
$\mathrm{x}_i(t)$ is the **current** covariate vector, updated each month. $\boldsymbol{\beta}$ is a single coefficient vector estimated across all loan-months.

Log partial likelihood (maximised to obtain $\hat{\boldsymbol{\beta}}$)

$$\ell(\boldsymbol{\beta}) \;=\; \sum_{i:\,\delta_i=1} \Bigg[\; \underbrace{\boldsymbol{\beta}^\top \mathrm{x}_i(t_i)}_{\substack{\text{log-score of}\\ \text{the prepaying loan}}} \;-\; \underbrace{\log \sum_{j\in\mathcal{R}(t_i)} \exp\!\big(\boldsymbol{\beta}^\top \mathrm{x}_j(t_i)\big)\Bigg]}_{\substack{\text{sum over all loan-months}\\ \text{active at } t_i}}$$

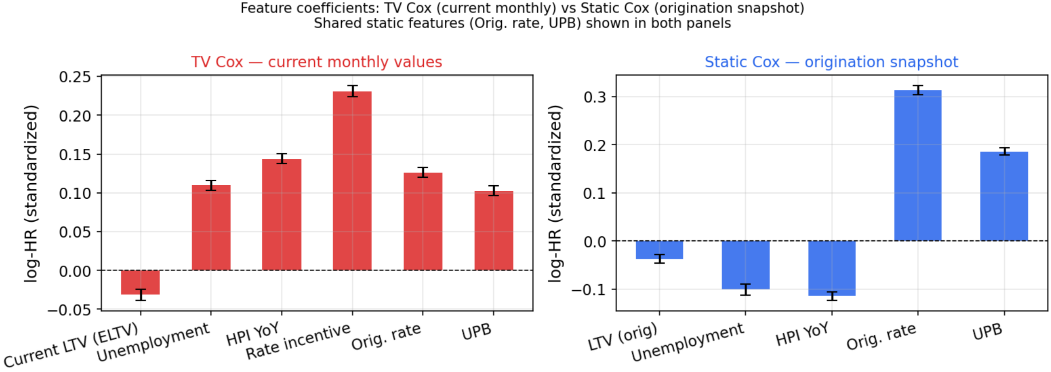
At each event: *given a prepayment at t_i, what is the probability it was loan i rather than any other active loan?* Maximising ℓ estimates $\hat{\boldsymbol{\beta}}$ without specifying $h_0(t)$.

Time-varying feature trajectories by vintage.



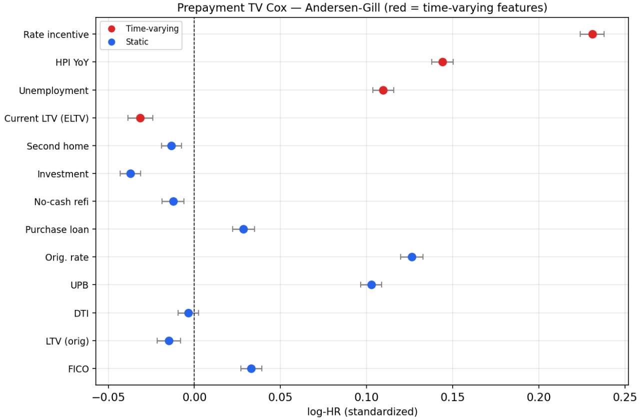
Static Cox uses only month-0 values — Andersen-Gill tracks these every month.

Static vs. TV Cox coefficient comparison.



- `rate_incentive`: dominant positive predictor in TV Cox; near zero at origination, absent from Static Cox
- Macro signals (unemployment, HPI) flip sign — vintage-era snapshots capture a different regime than monthly values

Landmark C-index: TV Cox leads to T=60, then reverses.



Landmark C-index

$$C(t^*) = P\big(\hat{h}_i(t^*) > \hat{h}_j(t^*) \mid T_i \le t^* < T_j\big)$$

Horizon	Static	**TV**	Δ
$T = 12$	0.609	**0.681**	+0.072
$T = 24$	0.627	**0.687**	+0.060
$T = 36$	0.630	**0.675**	+0.045
$T = 60$	0.625	**0.646**	+0.021
$T = 84$	**0.623**	0.605	−0.018
$T = 120$	**0.620**	0.583	−0.037

Key Takeaways: E(ii)

1. Monthly-updated covariates add real information ($+1{,}996$ log-likelihood gain).
2. **TV Cox leads at short/medium horizons:** $+0.072$ **at** $T{=}12$, $+0.021$ **at** $T{=}60$.
3. **TV Cox degrades beyond** $T{\approx}70$ **months — use a landmark super-model.**
4. Rate incentive is the key driver; macro coefficients require careful inspection.

Part E(iii)

Neural Survival Models: LongitudinalDeepHit

Transformer Encoder · Competing-Risk CIFs · Gradient Sensitivity

Why go beyond Cox?

Cox / Andersen-Gill limitations

- Proportional hazards: covariate effect is **constant over time**
- Single scalar score — cannot model the **S-curve shape** of prepayment
- Competing risks require separate model fits; no shared representation
- Cannot exploit the **sequence structure** of monthly covariate history

What a sequence model can do

- Read the full monthly path $[x_1, \ldots, x_T]$ — **burnout is encoded**
- Produce nonlinear, time-varying hazard rates $\lambda_k(t \mid x)$
- Model prepayment and default **jointly** via shared encoder
- Attention weights identify **which months** drive the prediction

Architecture rationale.

Alternatives rejected

- **LSTM** — vanishing gradients at $T=120$; attends only to adjacent months; hard to interpret
- **Single-context head** — no time index in head; near-constant hazard curve; gradient flows only through last hidden state
- **Two separate models** — violates independent censoring (defaulted loans are riskiest borrowers, not random exits); loses shared macro signal

Why this combination

- **Causal Transformer** — reaches any past month in one attention step; gradient flows through every position; weights are interpretable
- **Per-timestep head** — $\lambda(t)=\text{head}(h_t)$: hazard conditioned on full causal history $x_1 \ldots x_t$; every month trained independently
- **Joint encoder** — shared representation captures that the same macro shock raises default and suppresses prepayment; chain product enforces $\text{CIF}_1+\text{CIF}_2+S=1$

LongitudinalDeepHit architecture.

Architecture

Input	$(B,\ T_{\max}{=}120,\ D{=}13)$ padded sequences
Proj.	Linear $D \to d_{\text{model}}{=}64$
PE	Sinusoidal positional encoding on loan age
Enc.	$\times 2$ pre-norm layers, 4 heads, causal mask
Head 1	FC $64 \to 64 \to 1$ per h_t, Sigmoid $\Rightarrow \lambda_1(t)$
Head 2	FC $64 \to 64 \to 1$ per h_t, Sigmoid $\Rightarrow \lambda_2(t)$
CIFs	Competing-risk chain product

Competing-risk CIFs

$$\mathrm{CIF}_k(t) = \sum_{s=1}^{t} \lambda_k(s) \cdot S(s-1)$$
$$S(t) = \prod_{s=1}^{t} (1-\lambda_1(s))(1-\lambda_2(s))$$

Guarantees $\mathrm{CIF}_1 + \mathrm{CIF}_2 + S = 1$ at every t.

Training objective

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \alpha\,\mathcal{L}_{\text{rank}}, \quad \alpha = 0.2$$

NLL: discrete-time cause-specific likelihood
Ranking: soft pairwise concordance (DeepHit-style)

Params	76,290
Data	101K loans, 4.5M rows
Epochs	41 (best ep. 29)
LR	3×10^{-4}, cosine
Loss	$11.30 \to 11.28$
Rank	$1.97 \to 1.84$

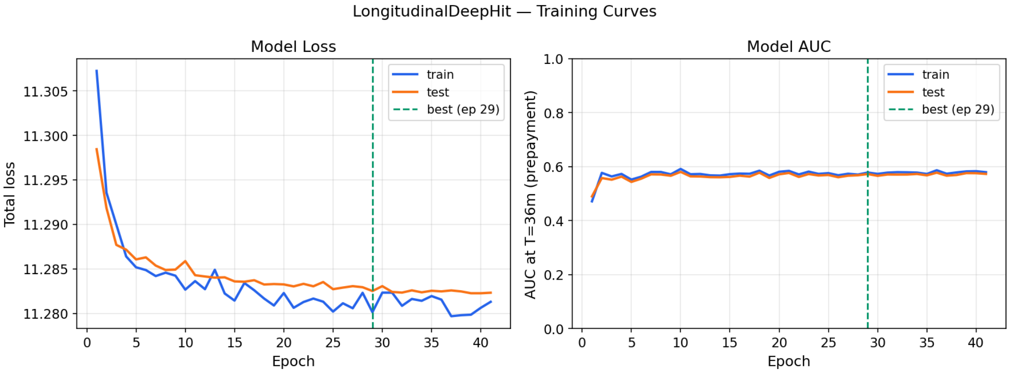
Data treatment

Each loan $\to$ 13-feature monthly sequence, zero-padded to $T_{\max}{=}120$ after exit.

Labels: prepayment $=1$, default $=2$, censored $=0$.

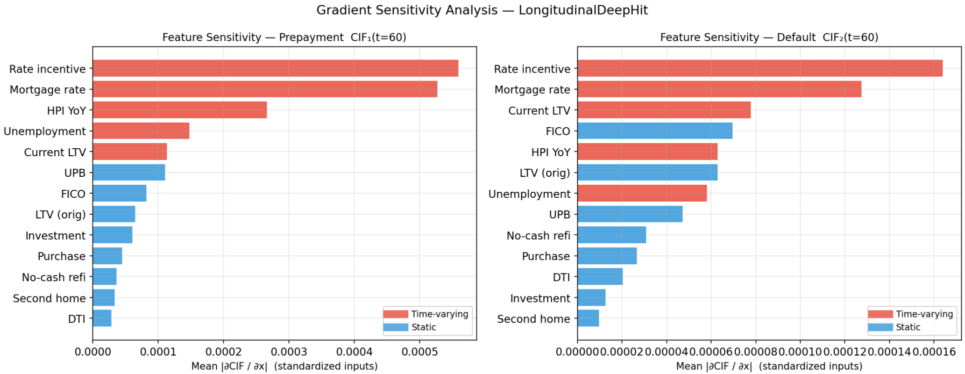
Causal mask; 80/20 train–test split.

Training dynamics: Negative Log Likelihood collapse and ranking loss.



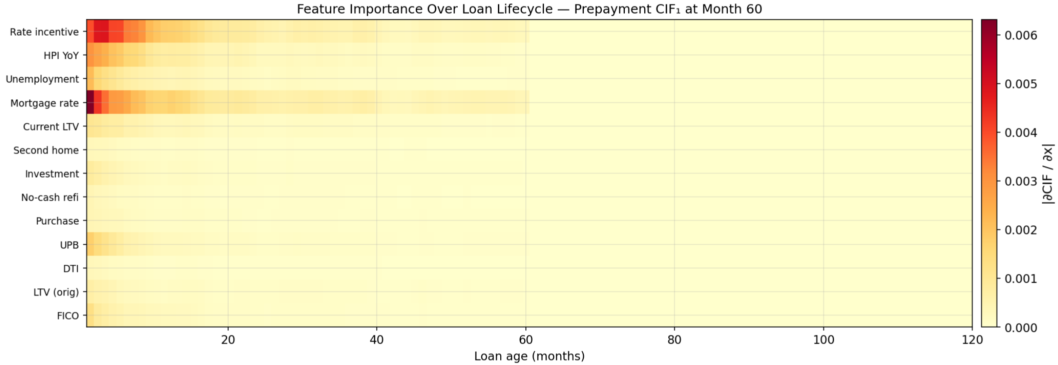
- **NLL freezes**: 1.6% event rate means predicting near-zero hazard for everyone already minimises the loss — rare-event gradients too weak to improve further.
- **Cox immune**: partial likelihood only ranks loans at event times, never predicts absolute hazard — no trivial solution under class imbalance.
- **AUC preserved**: ranking loss descends steadily, ordering loans by relative risk despite collapsed hazard values.

Gradient sensitivity by cause.



Red = time-varying features. Blue = static features.
Dynamic features rank above static for prepayment;
FICO and unemployment lead for default; consistent with cause-specific Cox in E(i).

Feature importance over the loan lifecycle.



`rate_incentive` and `Mortgage rate` dominates, especially in early months;

Model performance vs. DeepCox.

Prepayment CIF_1 — AUC at fixed horizons

Model	T=12	T=24	T=36	T=60
Part D DeepCox†	**0.684**	**0.701**	**0.721**	**0.729**
DeepHit (20K)‡	0.641	0.622	0.572	0.515

Default CIF_2 — AUC at fixed horizons

Model	T=12	T=24	T=36	T=60
Part D DeepCox†	0.684	0.701	**0.721**	**0.729**
DeepHit (20K)‡	**0.687**	**0.708**	0.707	0.726

† 597K canonical ‡ 20K hold-out green = better

Prepayment: DeepCox leads at all horizons (NLL collapse hurts long-horizon calibration).
Default: DeepHit leads at T=12 and T=24 with **6× fewer** training loans.

Key Takeaways: E(iii)

1. Default AUC matches DeepCox with $6\times$ fewer training loans.
2. Gradient sensitivity replicates Cox findings: rate incentive drives prepayment, FICO drives default.
3. Feature importance shifts over the loan lifecycle — unique to sequence models.
4. **Limitation: defaults are rare (1.6%), so the model learns to predict near-zero hazard for everyone — AUC is preserved but absolute probabilities are unreliable; fix via focal-loss upweighting.**

Part E(iv)

Scenario Analysis: MBS Prepayment under Rate Shocks

TV Cox Rate Channel · Survival Curves · Monte Carlo MBS Pricing

TV Cox: direct rate channel for scenario analysis.

Why TV Cox (E(ii)) is the primary model

`rate_incentive` $= r_{\text{orig}} - r_{\text{market}}(t)$ is a **direct time-varying covariate**.

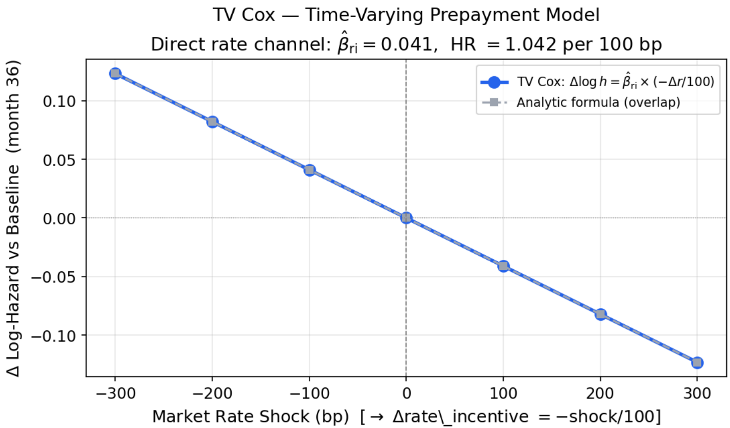
A Δr bp market rate shock immediately shifts the incentive by $-\Delta r/100$ pp, giving an analytic hazard response:

$$\Delta \log h(t) = \hat{\beta}_{\text{ri}} \times \frac{-\Delta r}{100}, \qquad \hat{\beta}_{\text{ri}} = 0.041 \text{ (HR} = 1.042 \text{ per 100 bp)}$$

The calibrated baseline hazard $\hat{H}_0(t)$ produces realistic $S(t)$, CPR, WAL, and price — no approximation needed.

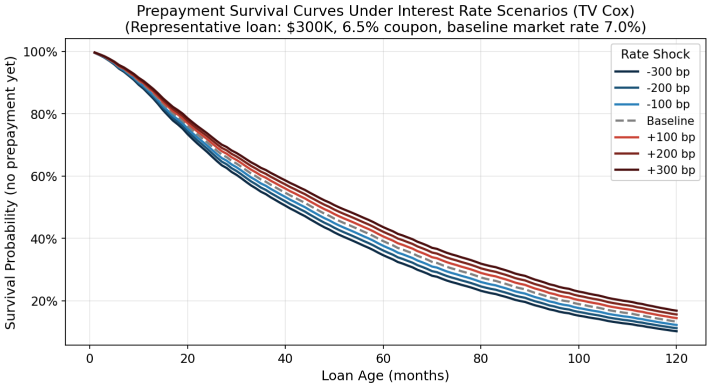
Representative loan: \$300K, 5.0% coupon, baseline market rate 5.3% (median, 1999–2025).
Scenarios: ± 100, ± 200, ± 300 bp — covering the BCBS standard stress range.

Shock sensitivity: TV Cox matches the analytic formula.



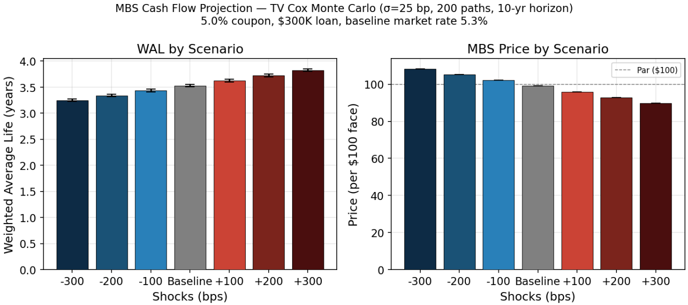
Each 100 bp rate cut increases the prepayment hazard by 4.2%; a 300 bp cut raises it by +0.123 in log terms. The model response matches the closed-form prediction exactly — no simulation needed, fully auditable.

Survival curves under seven rate scenarios.



Scenario	10-yr Survival	Median Prepay (mo.)
−300 bp	10.2%	41
−200 bp	11.2%	43
−100 bp	12.2%	45
Baseline	13.3%	46
+100 bp	14.4%	48
+200 bp	15.6%	50
+300 bp	16.8%	51

MBS: Weighted Average Life, MBS price, and negative convexity.



Scenario	WAL (yr)	Price ($)
−300 bp	3.25	108.32
−200 bp	3.34	105.26
−100 bp	3.43	102.17
Baseline	3.53	99.07
+100 bp	3.63	95.96
+200 bp	3.73	92.86
+300 bp	3.83	89.77

Negative convexity: when rates fall, borrowers prepay and return principal *at par*, capping price upside. Price gain (−300 bp: \$9.25) $<$ loss (+300 bp: \$9.30).

MC: 200 paths, σ=25 bp.

Key Takeaways: E(iv)

1. **TV Cox has a linear rate channel**: every 100 bp rate cut adds the same $+4.2\%$ to prepayment hazard — equal steps in rate produce equal steps in response, making results predictable and fully explainable.
2. **Negative convexity confirmed**: when rates fall, borrowers prepay at par, capping price upside ($\$9.25$ gain $<$ $\$9.30$ loss at ± 300 bp).

Thank you

Questions?

Andersen & Gill (1982) *Ann. Stat.* · Fine & Gray (1999) *JASA* · Lee et al. (2018) *DeepHit* · Sadhwani et al. (2021) *J. Fin. Econometrics* · BCBS (2019)